NG JIE YANG

Undergraduate - NUS Computer Science
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SUMMARY

NUS Computer Science undergraduate with experience across robotics fleet orchestration, ADAS perception, computer vision, and full-stack engineering. Strong focus on modern C++, Python, ROS, GPU-accelerated vision, backend workflow systems, and production validation for real-world robotics deployments.

EDUCATION

National University of Singapore | *Bachelor of Computer Science; NUS Merit Scholarship* 2024 - 2028 (Expected)
Anglo-Chinese School (Independent) | International Baccalaureate Diploma Programme, 42/45 (Top 10% Globally)

TECHNICAL SKILLS

Languages: C++, Python, Java, TypeScript, JavaScript, SQL
Robotics & Perception: ROS 2, NVIDIA VPI, OpenCV, AprilTags, YOLOv8, RGB-D cameras, camera calibration, surround-view stitching, optical flow, KLT tracking, BEV processing, ego-motion estimation
Machine Learning / GPU Computing: CUDA, PyTorch, NumPy, Hugging Face, Roboflow, feature detection & tracking, model evaluation
Backend & Systems: FastAPI, Node.js, REST APIs, Docker, asynchronous workflows, plugin architecture, workflow automation
Data / Messaging: SQLite, Redis, RabbitMQ, MQTT, Firebase
Frontend & Tools: React, Vite, Tailwind CSS, JavaFX, FXML, Git, GitHub Actions, Linux, Gradle

PROFESSIONAL EXPERIENCE

ST Engineering | *Software Engineering Intern - Computer Vision & Autonomous Systems* May 2026 – August 2026

- Developed a real-time ADAS perception module in modern C++ for BEV-based ego-motion estimation and under-vehicle view reconstruction, integrating NVIDIA VPI, CUDA, and OpenCV in a modular low-latency pipeline.
- Implementing feature-track management, motion estimation, and frame-warping logic to transform road-surface correspondences into vehicle translation and rotation estimates for continuous temporal stitching of downward-view imagery.
- Designed and implemented an automated multi-camera extrinsic calibration system using AprilTags, reducing production camera calibration time by over 90% (1 hour to under 5 minutes) through automatic camera pose estimation with minimal operator intervention.
- Built a GPU-accelerated surround-view stitching engine using dense optical flow, perspective warping, and multi-band blending to generate seamless panoramic video for multi-camera vehicle perception.
- Designed custom CUDA kernels, concurrent GPU execution pipelines, and optimized memory transfers for optical flow, image remapping, and blending, significantly reducing processing latency and enabling real-time performance on NVIDIA GPU platforms.

Movel AI | *Full-Stack Developer Intern - Robotics Fleet Orchestration* Dec 2025 - Apr 2026

- Developed backend services in **Node.js/TypeScript** for a robotics fleet orchestration platform coordinating autonomous mobile robots and enterprise systems.
- Designed and implemented **REST APIs**, workflow orchestration logic, and state management services for robot task execution and inventory operations.
- Built fault-tolerant workflows with validation, retries, resource reservation, and recovery mechanisms to maintain consistency across distributed robotics subsystems.
- Integrated backend services with **RFID systems, MQTT-based device communication, elevator controls, door access systems, and inventory management platforms**.
- Containerized and deployed services using **Docker**, supporting production deployments and operational validation at NUS and GSK.

SELECTED PROJECTS

AI-Assisted Ophthalmic Surgical Tool Identification System |
Industry-Partnered Computer Vision & Robotics Project - Singapore National Eye Centre Jan 2026 - Present

- Prototyping a perception-to-grasp pipeline for automated identification and sorting of ophthalmic surgical instruments in collaboration with Singapore National Eye Centre.
- Evaluating YOLOv8, depth-camera sensing, and vision-language model approaches for fine-grained recognition of reflective, visually similar, and partially overlapping surgical tools.
- Integrated RGB-D sensing with ROS-based robot control to map detected instrument locations into grasp coordinates for downstream UFactory robotic arm manipulation.

Gamified Guitar Learning Application | *Software Engineer & Co-Lead - NUS Orbital Artemis Award* May 2025 - Aug 2025

- Developed a real-time computer vision guitar learning app using YOLOv8 detection and calibration algorithms, achieving 90% note-overlay accuracy at 30 FPS.
- Integrated Web Audio API pitch detection for real-time audio feedback, enabling synchronized play-along features, timing evaluation, scoring, and gamified learning tools.

ADDITIONAL EXPERIENCE

Inland Revenue Authority Singapore | Tax Officer and Customer Service Assistant | Jan 2024 - Jun 2024 - Resolved taxpayer queries across income tax types and promoted adoption of IRAS e-services and chatbot support channels.

Robotics Club | 2016 – 2019 Competing Team | Designed, coded, and built robots based on given guidelines and objectives in different competitions | Selected to represent school at the Singapore Vex Robotics Championship 2016 to 2019 (1st Individual and Team Placing), the VJC Robotics Competition 2017 to 2019 (1st Team Placing), the Singapore VRC National Championship 2019 (1st Team Placing) and MakeX Singapore (2nd Team Placing)